Problem 2

What sequence of states best explains a sequence of observations ?

$$O = \{O_1, O_2, \dots, O_T\} \qquad Q = \{q_1, q_2, \dots, q_T\}$$

Theory

What does best mean ?

1. Choose states that are individually most likely

$$\gamma_t(i) = P(q_t = S_i \mid O, \lambda)$$
$$= \frac{\alpha_t(i)\beta_t(i)}{P(O \mid \lambda)} = \frac{\alpha_t(i)\beta_t(i)}{\sum_j \alpha_t(j)\beta_t(j)}$$

Therefore, $\sum_i \gamma_t(i) = 1$ and $q_t = \underset{i}{\operatorname{argmax}} (\gamma_t(i))$

Will maximise expected number of correct states but Q might not make sense

2. Choose a path sequence that maximises $P(Q \mid O, \lambda) = P(Q, O \mid \lambda)$

Using Viterbi algorithm :

$$\delta_t(i) = \max_{q_1, q_2, \dots, q_{t-1}} P(\{q_1, q_2, \dots, q_t = i\}, \{O_1, O_2, \dots, O_t\} \mid \lambda)$$

With : $\delta_1(i) = \pi_i \, b_i(O_1)$

$\psi_1(i) = 0$

Induction: $\delta_{t+1}(j) = \max_i (\delta_t(i)\, a_{ij})\, b_j(O_{t+1})$

$\psi_{t+1}(j) = \underset{i}{\operatorname{argmax}} (\delta_t(i)\, a_{ij})$

Termination: $P^* = \max_i (\delta_T(i))$

$q_T^* = \underset{i}{\operatorname{argmax}} (\delta_T(i))$